

L_RAG: A Controlled Ablation Study of Vietnamese Legal Retrieval-Augmented Generation

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Abstract

Legal question answering must retrieve evidence that is relevant, complete in its statutory context, valid at the relevant time, and traceable to an authoritative source. We evaluate L_RAG, a Vietnamese legal retrieval-augmented generation pipeline that combines E5 dense retrieval, BM25, a knowledge-graph overlay, reciprocal-rank fusion, cross-encoder reranking, and citation-grounded generation. On a fixed 500-question benchmark (400 answerable and 100 unanswerable questions), we report five controlled ablation families rather than a pooled factorial comparison: embedding text, dense versus hybrid retrieval, graph expansion, reranking within a shared candidate pool, and Base versus chain-of-thought (CoT) prompting. Metadata improves document success at depth 10 by 4.5 percentage points but reduces MRR. The tested BM25+RRF hybrid loses 5.98 Recall@10 points to dense retrieval and increases median total latency from 5.483 to 31.538 seconds. One-hop graph expansion loses 11.81–12.40 Recall@10 points, chiefly by replacing seed evidence. In a common 30-candidate pool, Jina reranking improves Recall@10 by 9.23 points and nDCG@10 by 11.85 points over RRF-only. Under fixed context, CoT decreases GPT-4o-mini Token F1 by 1.57 points. Retrieval is scored against gold chunk identifiers; lexical answer overlap, template-based answerability behavior, structural citation fields, and recorded latency are separate diagnostics. These are benchmark-bounded engineering findings: they do not establish legal correctness, citation entailment, or faithful reasoning.

1 Introduction

Legal information systems must do more than return text that appears related to a question. A useful answer must be grounded in an authoritative provision, retain the qualifiers and exceptions that shape its meaning, and respect the provision’s temporal status. These requirements are especially de-

manding for Vietnamese legal corpora, where a citizen’s everyday wording may differ sharply from statutory language and where an answer can depend on a provision’s place in a legal hierarchy. This report combines the system, dataset, protocol, and ablation evidence into one traceable account of L_RAG.

1.1 Background and Motivation

Retrieval-Augmented Generation (RAG) gives a language model access to external evidence rather than relying solely on model parameters. In a legal setting, however, a semantically similar passage is not automatically the governing rule. A retrieved answer may omit an exception, point to a repealed instrument, or cite a document without identifying the article and clause that support the conclusion. The system therefore needs to balance semantic matching, exact terminology, document structure, and evidence ordering before generation begins.

For example, a Vietnamese question about whether a land-use transaction is permitted may use everyday wording, while the controlling provision uses a formal land category, an effective date, and a cross-reference to another article. A keyword-only search can miss the paraphrase; a vector-only search can retrieve a broad discussion; and a generator without source constraints can produce a fluent but unsupported explanation. L_RAG treats the answer as an evidence assembly problem: identify the source document, recover the relevant provision and neighboring context, rank the available chunks, and show the citations used to form the response.

1.2 Problem Statement

The study addresses five recurring failure modes in legal RAG:

1. **Vocabulary mismatch.** Users describe a legal problem in everyday Vietnamese, whereas statutes use formal terms, identifiers,

and numeric thresholds. Dense retrieval helps with paraphrase, while lexical retrieval can preserve exact legal terminology.

2. **Lost statutory context.** Laws are hierarchical—from a legal instrument to chapters, articles, clauses, and items. Fixed-size chunks can separate a rule from its scope, definition, or exception.
3. **Cross-document dependencies.** Provisions can cite, amend, replace, or repeal one another. A relevant answer may require a relation edge in addition to a similar passage.
4. **Temporal and authority errors.** Semantic similarity does not determine whether an instrument was effective at the requested date or whether a lower-authority rule has been superseded.
5. **Unsupported answers.** A model can answer confidently when the retrieved context is incomplete. Citation markers and refusal strings must be measured separately from legal correctness.

L_RAG combines dense and sparse retrieval with a graph overlay, a shared candidate-pool reranker, and citation-oriented prompting. The central question is not whether every component is plausible in isolation, but which components improve the evidence available to the generator under a controlled evaluation protocol.

1.3 Project Objectives

The project has four operational objectives:

- build an end-to-end Vietnamese legal RAG pipeline using dense E5 retrieval, BM25, reciprocal-rank fusion (RRF), graph expansion, reranking, and citation-grounded generation;
- represent legal hierarchy, reading order, and document relationships in a query-time knowledge-graph overlay;
- compare embedding text, retrieval mode, graph policy, reranker, and generator prompt choices with question-level pairing; and
- evaluate retrieval, answer-overlap, abstention, citation-structure, and latency behavior on a 500-question Vietnamese legal benchmark.

Each objective has an observable artifact. The corpus and benchmark are described in Section 3; the online and offline modules are specified in Section 4; frozen manifests and metric definitions are given in Section 5; and the five completed comparisons, with inline links to their evidence, are reported in Section 6.

1.4 Scope of the System

The evaluated scope includes Vietnamese legal-document ingestion, normalization, provision and chunk construction, relationship processing, vector and BM25 indexing, graph-based filtering or expansion, candidate reranking, prompt serialization, generation, and post-hoc scoring. The benchmark uses stored question IDs, reference answers, and gold document, provision, and chunk identifiers. The report consequently evaluates a frozen research pipeline and its saved outputs rather than a live legal service.

The scope excludes autonomous legal advice, formal legal interpretation, claim-level citation-entailment verification, expert adjudication of legal correctness, and guaranteed coverage of every current external instrument. The corpus is a snapshot; unresolved external references are represented as stubs, and graph direction or validity metadata can require further review. The generator is instructed to abstain on insufficient context, but an abstention marker is only an observable output diagnostic. These boundaries are part of the system definition, not post-hoc qualifications.

1.5 Main Contributions

This report contributes a controlled empirical account of L_RAG rather than an unqualified claim of legal reliability:

1. a structured Vietnamese legal corpus that preserves document, provision, chunk, metadata, and relationship provenance;
2. a G-LRAG graph overlay for containment, reading order, legal relations, authority, and validity-aware query filtering;
3. an end-to-end retrieval and generation pipeline combining E5 dense retrieval, optional BM25/RRF, graph expansion, cross-encoder reranking, and cited output formatting; and
4. five paired ablation families that distinguish document discovery from exact evidence

ranking, isolate graph and reranker effects, and separate observable prompt behavior from legal correctness.

The most important methodological contribution is the separation of evidence availability, answer overlap, refusal behavior, citation structure, and efficiency. A component is not called an improvement merely because one aggregate score increases: candidate ceilings, denominators, pairing, and claim boundaries are reported alongside every headline result.

1.6 Report Organization

The report follows a Problem → Data → Methodology → Implementation → Ablations → Discussion → Conclusion flow. Section 2 reviews RAG, legal information retrieval, hybrid ranking, GraphRAG, and evidence-grounded generation. Section 3 describes the corpus, graph-ready records, benchmark construction, and gold evidence. Section 4 turns those data products into the L_RAG architecture and query workflow. Section 5 defines frozen artifacts, metrics, pairing, uncertainty, and configurations. Section 6 reports the five controlled families with clickable table and figure references in the result sentences. Section 7 interprets trade-offs and sets safety boundaries, while Section 8 summarizes the evidence and prioritizes follow-up experiments. Appendices record schemas, examples, configuration details, observable prompt formats, and artifact pointers.

2 Background and Related Work

This section positions L_RAG at the intersection of retrieval-augmented generation, legal information retrieval, hybrid ranking, knowledge-graph retrieval, and citation-grounded generation. The purpose is not to claim that one paradigm solves legal QA by itself, but to make explicit which prior ideas motivate each component and why the report tests them separately.

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) supplements a language model’s parameters with documents retrieved at query time. A standard pipeline embeds a query, retrieves passages from an indexed corpus, serializes the top passages into a prompt, and asks a generator to produce an answer. The external context can reduce dependence on stale parametric knowledge

and gives a user sources to inspect, but it does not guarantee that the selected passages are sufficient or that the generated claims follow from them.

The basic RAG abstraction is useful for L_RAG because it separates three possible failure locations: the retriever can miss the governing passage, the context builder can omit its parent or exception, and the generator can fail to use or cite the evidence. Advanced methods such as Self-RAG (Asai et al., 2024) and Corrective RAG (Yan et al., 2024) add self-reflection or retrieval correction. They motivate measuring evidence availability and abstention, but they do not by themselves encode the formal hierarchy, amendment history, or temporal validity of Vietnamese legal instruments.

2.2 Legal Information Retrieval

Legal Information Retrieval (Legal IR) is a specialized area of information retrieval and legal informatics (Locke and Zuccon, 2022). Legal texts differ from open-domain documents in ways that change both indexing and evaluation:

1. **Hierarchical structure.** An instrument is organized into parts, chapters, articles, clauses, and items. The meaning of a chunk can depend on its parent heading or a neighboring exception.
2. **Precision requirements.** A missing qualifier, number, or negation can change the interpretation of a rule even when the retrieved paragraph is topically correct.
3. **Authority and validity.** The Constitution, laws, decrees, decisions, and circulars occupy different authority levels, and the applicable instrument depends on effective dates and amendment status.
4. **Explicit dependencies.** A provision may cite, amend, replace, or repeal another instrument. These relationships are evidence about context, not ordinary lexical similarity.

Legal-BERT (Chalkidis et al., 2020) and Lawformer (Xiao et al., 2021) show how domain-adapted transformers can represent legal language. Such encoders improve semantic matching, but a vector alone does not state whether an article is current, whether a reference is authoritative, or which sibling chunk supplies a missing exception.

L_RAG therefore retains structured identifiers and graph relations alongside the embedded text.

2.3 Dense, Sparse, and Hybrid Retrieval

Dense bi-encoders are effective for paraphrase and vocabulary variation, whereas sparse methods such as BM25 are strong at exact codes, names, numbers, and technical terminology. E5 (Wang et al., 2022) motivates the dense component in L_RAG; BM25 supplies a lexical complement when a question uses a statutory phrase verbatim. The trade-off is empirical: adding a modality can also change candidate ordering, tokenization behavior, and latency.

Reciprocal Rank Fusion (RRF) (Cormack et al., 2009) combines ranked lists without requiring calibrated scores. For a candidate d and retrieval systems $m \in M$,

$$\text{RRF}(d) = \sum_{m \in M} \frac{1}{k + r_m(d)}, \quad (1)$$

where $r_m(d)$ is the rank assigned by system m and k is a smoothing constant. L_RAG uses $k = 60$ in its RRF configurations. Because fusion can help one query while harming another, the ablation reports paired recall, rank, and latency deltas rather than assuming that hybrid retrieval is an unconditional improvement.

2.4 Knowledge-Graph-Augmented Retrieval

Knowledge graphs represent entities and explicit relationships as triples (e_1, r, e_2) . GraphRAG (Edge et al., 2024) shows how graph structure can support query-focused retrieval and multi-hop context. In a legal corpus, a graph can connect a document to its provisions, a provision to its chunks, adjacent chunks in reading order, and one instrument to another through citation, amendment, replacement, or repeal edges.

This structure can restore context that a fixed-size vector chunk omits. It also introduces failure modes that do not occur in a simple vector index: incorrect edge direction, stale validity fields, and a traversal policy that replaces high-confidence seeds under a fixed context budget. L_RAG’s G-LRAG overlay therefore applies validity and authority filters at query time and tests one-hop expansion as a distinct ablation. The graph result is interpreted as evidence about the tested policy, not about graph retrieval in general.

2.5 Reranking and Citation-Grounded Generation

Cross-encoder rerankers score the query and candidate passage jointly rather than comparing two independently encoded vectors. Passage reranking with BERT (Nogueira and Cho, 2019) motivates this second-stage design: it can improve the order of evidence at the cost of additional inference time. L_RAG compares multilingual backends including BGE-Reranker-v2-m3 (Chen et al., 2024), Jina Reranker v2 (Jina AI, 2024), and Qwen3 Reranker (Qwen Team, 2025) over a shared 30-candidate pool. Holding the pool fixed makes a positive effect interpretable as evidence promotion rather than new evidence access.

Citation-grounded generation adds source identifiers or provision references to the answer format and asks the generator to abstain when context is insufficient. These controls improve inspectability, but a citation marker is not the same as claim-to-source entailment. The report consequently treats citation presence, structural coverage, and refusal markers as observable diagnostics and does not convert them into legal-correctness labels.

2.6 Design Requirements for Legal RAG

The related work implies six requirements for a defensible legal RAG evaluation:

1. **Provenance:** every passage must retain document, provision, and chunk identifiers so a result can be inspected;
2. **Context preservation:** hierarchy and reading order must be available without assuming that a single chunk is self-contained;
3. **Validity awareness:** temporal status, authority, and relationship direction must be explicit and auditable;
4. **Controlled comparisons:** each ablation must freeze its benchmark, candidate pool, context budget, and generator conditions;
5. **Separate diagnostics:** retrieval coverage, lexical overlap, abstention, citation structure, and efficiency must not be collapsed into one unsupported quality score; and
6. **Safe interpretation:** automatic metrics must be reported with the missing human labels and must not be presented as legal advice.

L_RAG implements these requirements through structured corpus records, the G-LRAG overlay, frozen per-question artifacts, and paired analyses. The dataset and benchmark that make the requirements testable are described next.

3 Legal Dataset and Benchmark Construction

The experiments use two linked data products: a processed Vietnamese legal corpus and a question-answer benchmark with explicit evidence provenance. The ingestion path preserves the hierarchy of an instrument instead of flattening all text into unrelated passages. This section records the transformations, quality gates, and final statistics needed to reproduce the evaluation denominators.

3.1 Legal Corpus and Raw Data Sources

The corpus contains Vietnamese legal instruments such as National Assembly laws, government decrees, Prime Minister decisions, and ministry circulars. Each source record is associated with a stable document identifier, title, issuing authority, issue and effective dates, validity status, document type, and the raw body from which provisions and chunks are derived. Relationship records retain their source and target identifiers, relation type, and any available date or confidence metadata.

The processed inventory contains 151,624 documents, 1,386,267 provisions, 1,513,376 retrieval chunks, 883,256 document-level relationships, and 19,763 external-reference stubs. The scale summary is linked in [Table 1](#). The counts refer to processed records, not a claim that every current Vietnamese instrument is present or legally valid.

3.2 Dataset Normalization and Cleaning

Normalization converts heterogeneous source records into deterministic fields. Identifiers are canonicalized before joins; dates are represented in one machine-readable convention; document type and status values are mapped to a controlled vocabulary; and empty or malformed body fields are retained in an audit stream instead of silently becoming empty passages. The normalization rules are summarized in [Table 2](#).

Cleaning is intentionally conservative. A record is not discarded merely because it has an unresolved relation, since that relation can be useful as a visible limitation. Records with unusable identifiers, missing structural parents, or failed parsing

gates are quarantined and counted separately from the accepted corpus.

3.3 Relationship Processing and External References

Relationship processing separates resolved in-corpus targets from references that point outside the snapshot. Resolved edges store a canonical source, target, relation vocabulary value, direction, and provenance. An unresolved target becomes an `ExternalStub` with the original reference string; the stub is not treated as retrievable legal text. Direction is checked against the relation semantics before a graph edge is made available to query-time traversal.

The relation inventory supports both structural context recovery and legal dependency analysis. It does not turn every citation into a validity decision: authority and effective-date overlays are applied only when the relevant fields and edge direction have passed verification.

3.4 Quarantine and Data-Quality Verification

Quality verification runs before a record contributes to the retrieval index or graph. The gates check identifier uniqueness, document-to-provision and provision-to-chunk parentage, non-empty indexed text, relation endpoint resolution, and direction consistency. A failed gate writes the record and reason to a quarantine ledger so that the final inventory can be reconciled without guessing how many records were removed.

This quarantine policy is conservative by design. It avoids propagating an incorrect legal edge, but it can lower recall by excluding records whose metadata is incomplete. That trade-off is part of the corpus limitations discussed in [Section 7](#).

3.5 Legal Text Structuring

The canonical hierarchy is *Document* → *Provision* → *Chunk*. A document stores source-level metadata; a provision stores article, clause, item, and heading information; and a chunk stores the text presented to the retriever, its sequence number, and pointers back to both parents. The embedded text can be chunk-only or metadata-enriched, but the provenance fields remain present in both variants.

For each indexed passage, the minimum prove-

Table 1: Processed Vietnamese legal corpus inventory.

Entity	Description	Count
Documents	Full legal instruments and source metadata	151,624
Provisions	Articles, clauses, items, and hierarchical units	1,386,267
Chunks	Retrieval passages with parent pointers	1,513,376
Document relationships	Amend, replace, cite, repeal, and related links	883,256
External stubs	Referenced entities not resolved in the corpus	19,763

Table 2: Normalization rules applied before indexing.

Field	Rule and audit behavior
Identifier	Trim, canonicalize separators, and preserve the source key for traceability.
Dates	Parse issue/effective dates into a consistent date representation; retain missing values for audit.
Document type	Map source labels to a controlled type vocabulary; keep the raw label as metadata.
Status	Preserve effective, amended, repealed, and unknown states rather than inferring validity from text.
Text	Normalize whitespace and headings while retaining the original body and provision order.
Relations	Resolve in-corpus targets; create an external stub when the target is unavailable.

Table 3: Relationship vocabulary used by the graph-ready corpus.

Relation	Interpretation
DOCUMENT_HAS_PROVISION	Document contains a hierarchical provision.
PROVISION_HAS_CHUNK	Provision owns an indexed passage.
CHUNK_NEXT	Adjacent chunks in reading order.
PROVISION_NEXT	Adjacent provisions in reading order.
CITES	Source instrument explicitly cites a target instrument.
AMENDS/REPEALS	Source changes or terminates a target instrument.
REPLACES	Source supersedes a target instrument.

nance object is

ChunkMeta = {chunk_id, doc_id,
provision_id, title,
issuing_authority, effective_date,
sequence}.

(2)

This structure lets evaluation match retrieved IDs to gold evidence even when the visible passage is shortened for a prompt. It also lets the graph restore neighboring passages without losing the source document.

3.6 QA Benchmark Construction

The benchmark contains 500 Vietnamese legal questions: 400 answerable questions with gold evidence and 100 deliberately unanswerable questions. Candidates were generated from bundles of approximately three to five related provisions and chunks using `gemini-3.1-flash-lite`. A

Table 4: Data-quality gates and their downstream effect.

Gate	Action on failure
Identifier and parent integrity	Quarantine; do not emit a retrievable child record.
Text availability	Quarantine empty or unparseable passages; retain audit reason.
Relation endpoint	Resolve target or emit an external stub.
Relation direction	Keep in audit stream until direction is verified.
Validity metadata	Preserve unknown status and exclude it from strict filters.
Count reconciliation	Block export until document/provision/chunk counts agree.

separate verification pass checked answerability from the supplied evidence, absence of unsupported claims, category and difficulty labels, and consistency of the reference answer. Diversity filtering and deduplication were applied only after candidates passed those checks.

The final generation run sampled 515 documents and 1,697 provisions, and 72.41% of candidates passed verification before filtering. The resulting question records carry the same stable `qa_id` through retrieval, generation, and post-hoc analysis. Unanswerable questions have empty gold evidence by construction so that refusal behavior can be measured separately from answer generation.

3.7 Question Categories and Answer Types

Question categories encode the evidence pattern rather than the expected answer length. Single-hop questions can be answered from one local passage; citation questions require identifying a named authority; legal-validity questions depend on status or date; multi-hop questions require more than one linked passage; and cross-document questions require a relation between instruments. Answer types are abstractive, boolean, extractive, or unanswerable. The final category counts are linked in Table 5.

The category counts sum to 500, while the answer-type counts intentionally separate the 400 answerable records from the 100 unanswerable

Table 5: Benchmark category and answer-type inventory.

Category	Count	Answer type	Count
Single hop	265	Abstractive	199
Citation	106	Boolean	123
Legal validity	96	Extractive	78
Multi hop	21	Unanswerable	100
Cross-document	12	Answerable total	400

records. Small subgroups, particularly multi-hop and cross-document questions, are retained for diagnosis but are not treated as stable estimates of generalization.

3.8 Ground-Truth Evidence

Each question stores its text, category, answer type, reference answer, and a ground-truth object containing the supporting document, provision, and chunk identifiers:

$$Q_i = \{\text{qa_id, question, category, answer_type, reference_answer, ground_truth}\}. \quad (3)$$

For answerable questions, the gold chunk set G_i is the retrieval relevance unit used for $\text{Recall}@k$, $\text{Hit}@k$, $\text{MRR}@k$, and $\text{nDCG}@k$. The document and provision IDs are retained so that document-level success and structural coverage can be analyzed alongside exact chunk ranking. For unanswerable questions, $G_i = \emptyset$ by design; they are not silently inserted into retrieval denominators.

Gold evidence is an annotation target, not a complete legal theory. A legal expert may regard an alternative provision or a broader authority set as sufficient, and the benchmark does not provide that semantic judgment. The evaluation therefore states exactly what the gold IDs measure and what they do not establish.

3.9 Dataset Statistics and Validation

Before export, the pipeline reconciles source, normalized, structured, and indexed counts. Every document points to zero or more provisions; every indexed chunk points to one provision and one document; every resolved legal edge points to existing endpoints; and every external reference points to a stub rather than an invented target. Missing-text, orphan, quarantine, and unknown-validity counts are retained in the audit ledger even when the final retrieval index excludes them.

The validation identity is checked at each stage:

$$\begin{aligned} N_{\text{input}} &= N_{\text{accepted}} + N_{\text{quarantined}}, \\ N_{\text{relations}} &= N_{\text{resolved}} + N_{\text{external}}. \end{aligned} \quad (4)$$

These identities make the corpus summary in [Table 1](#) auditable. They also explain why a graph or strict validity filter can reduce available evidence: the system prefers a visible missing or uncertain record to an unverified relation presented as authoritative.

4 L_RAG System Design and Methodology

L_RAG turns the structured corpus and benchmark into an evidence-oriented retrieval and generation pipeline. Offline processing creates normalized records, graph artifacts, and vector or sparse indexes. Online processing resolves query scope, retrieves candidate chunks, optionally expands or filters them with the graph, reranks a bounded pool, and serializes the final context for a citation-grounded generator. The same document, provision, and chunk identifiers introduced in [Section 3](#) are carried through every stage.

4.1 End-to-End System Architecture

The architecture has five operational phases: (1) query and temporal-scope resolution, (2) dense and optional sparse retrieval, (3) graph filtering or context expansion, (4) rank fusion and candidate reranking, and (5) prompt construction, generation, and cited output. [Figure 1](#) shows the data flow and the point at which each family of ablations changes a component.

The component boundaries are deliberate. The retriever decides which chunks are available, the graph can alter that set under a fixed context budget, the reranker only reorders a candidate pool, and the generator sees a serialized top-10 context. This separation prevents a generation metric from being mistaken for evidence-retrieval quality.

4.2 Offline Data and Index Construction

Offline construction branches from the normalized records into three linked artifacts. The structured-text branch emits provisions and chunks with stable parent pointers. The graph branch emits document, provision, chunk, and relationship nodes plus validity and authority metadata. The retrieval branch embeds either chunk-only or header-plus-chunk text into a FAISS index and builds tokenized

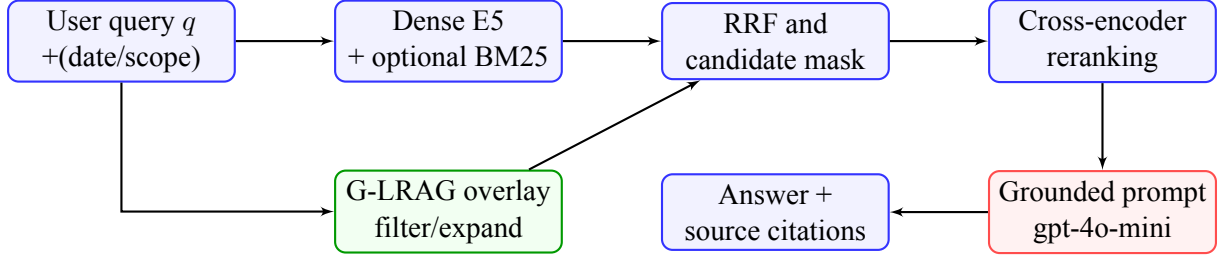


Figure 1: End-to-end operational workflow of L_RAG. The graph and reranker stages are optional branches whose effects are tested under separate frozen ablation controls.

BM25 shards for configurations that enable sparse retrieval.

The dense index uses L2-normalized vectors and inner-product search, which is equivalent to cosine ranking for normalized embeddings. The graph and index artifacts keep the same chunk IDs, so a retrieved vector row can be joined back to its document and provision without fuzzy matching. Frozen manifests record the corpus snapshot, model checkpoint, index path, configuration name, and benchmark version used to create each evaluation run.

4.3 Online Query-Processing Workflow

At query time, the system first preserves the raw Vietnamese question and any explicit date or scope constraint. The query encoder produces an E5 vector; the dense branch retrieves a broad seed set, while the optional BM25 branch searches tokenized shards. The branches are merged with RRF when hybrid mode is enabled. A graph configuration can then apply an ID whitelist or expand five seeds by one hop along structural edges. The resulting candidates are trimmed to the family-specific budget, reranked when enabled, and serialized as the final ten chunks.

Insufficient evidence follows an observable fallback path. If no candidate survives a valid scope filter, if the context contains no usable text, or if the prompt’s evidence threshold is not met, the generator is instructed to emit `INSUFFICIENT_CONTEXT`. This marker is saved for evaluation; it is not treated as proof that the question was legally unanswerable. Invalid filters and unresolved external references remain visible in the run record instead of being silently repaired.

4.4 Knowledge Graph Construction

The G-LRAG graph uses Document (D), Provision (P), Chunk (C), and ExternalStub nodes. Structural edges

encode `DOCUMENT_HAS_PROVISION`, `PROVISION_HAS_CHUNK`, `CHUNK_NEXT`, and `PROVISION_NEXT`. Legal-relation edges encode citations, amendments, replacements, and repeals. Each edge retains source provenance and is available to query-time traversal only after direction and endpoint checks.

The overlay folds validity fields relative to an `as_of_date` parameter and can sort candidates by authority rank. A simplified precedence ordering used for filtering is

$$\text{Constitution} \succ \text{Law/Code} \succ \text{Decree} \succ \text{Circular}. \quad (5)$$

Reading-order traversal follows adjacent chunks or provisions to recover nearby context. In the completed graph ablation, however, the traversal is limited to five seeds and at most ten retained chunks; this policy can replace strong seed evidence and is therefore evaluated as a change to the candidate set, not as a free context addition.

4.5 Retrieval and Ranking

The dense encoder is `intfloat/multilingual-e5-large`. For the metadata condition, the indexed string is

$$\text{Text}_{\text{embed}} = \text{DocumentTitle} \parallel \text{ProvisionHeader} \parallel \text{ChunkContent}. \quad (6)$$

the chunk-only condition removes the first two fields. The FAISS index uses `IndexFlatIP`. Sparse retrieval uses `rank_bm25` with `underthesea` tokenization and $k_1 = 1.5$, $b = 0.75$. Dense and sparse lists are fused by

$$\text{RRF}(d) = \sum_{m \in \{\text{dense}, \text{BM25}\}} \frac{1}{60 + r_m(d)}. \quad (7)$$

The dense/hybrid/graph family retrieves 50 seeds and retains at most ten final chunks. The reranker family reconstructs a common RRF pool

of 30 candidates (C_{30}), then retains ten (T_{10}). mMiniLM, Qwen3, BGE, and Jina score the full query–candidate pair with a cross-encoder. Models are loaded, executed, and unloaded sequentially to avoid GPU memory interference. The reranker can improve ordering inside C_{30} , but it cannot recover evidence that never entered the pool.

4.6 Answer Generation

The context serializer writes each retained chunk with its chunk ID, document title, provision header, and passage text. The generator receives a fixed temperature of 0.0, a maximum of 1,024 output tokens, and instructions to answer in the requested type: abstractive, boolean, extractive, or unanswerable. A citation should identify the governing article or instrument, and an insufficient context should produce the visible marker `INSUFFICIENT_CONTEXT`.

The observable output record contains the final answer, citation markers, parser fields, refusal marker, latency, and the ordered context IDs. It does not contain hidden chain-of-thought, reasoning-token accounting, or a human claim-entailment annotation. Consequently, the evaluation can report output format and lexical overlap but cannot infer legal correctness from a fluent answer or a well-formed citation.

4.7 End-to-End Worked Example

Consider a benchmark question asking whether a specified land-use action is permitted under a named Vietnamese instrument. The source record is first joined to its document ID and effective date. Its controlling article is represented as a provision, and the provision is split into ordered chunks whose IDs remain linked to the parent. The query encoder retrieves candidate chunks; the graph can add the adjacent clause or a cited instrument; RRF and the reranker order the retained candidates; and the serializer presents the top ten with their provenance.

The generator then produces an answer in the requested format and attaches the document and provision identifiers used as citations. If the retrieved context contains only a related definition but not the operative condition, the intended output is `INSUFFICIENT_CONTEXT`. The example illustrates why the report distinguishes document success from exact chunk recall and citation presence from citation entailment: each is a different observable stage in the same traceable workflow.

5 Implementation and Experimental Setup

This section states the controls and scoring rules needed to interpret the ablations in Section 6. The benchmark and corpus are shared where possible, but the experiment families are not one factorial experiment: candidate pools, context construction, and latency accounting differ. A result is therefore compared only with its paired control inside the same family.

5.1 Pipeline and Frozen Artifacts

The evaluated pipeline has five stages: query embedding with `intfloat/multilingual-e5-large`; dense FAISS retrieval and, where enabled, tokenized BM25 retrieval; RRF fusion, graph expansion, or cross-encoder reranking; serialization of retained chunks into a grounded prompt; and answer generation and evaluation. The analysis reads frozen manifests, predictions, aggregate metrics, latency records, and error logs; it does not rerun models.

Every primary comparison is joined by `qa_id`. A valid row has a prediction, no saved error, and no failed or skipped status. The same ID is used to join the question category, reference answer, gold chunk set, ordered retrieved context, output diagnostics, and latency. This row-level alignment is what makes a reported mean difference traceable to paired cases.

The completed work is organized around five research questions:

1. Does document and provision metadata improve exact evidence retrieval or document-level coverage?
2. Does dense–sparse hybrid retrieval improve coverage or rank over dense retrieval alone?
3. Does the current G-LRAG graph policy improve evidence coverage or downstream behavior under a ten-chunk context budget?
4. Which reranker best trades ranking quality against cost when the candidate pool is fixed?
5. Does an explicit chain-of-thought prompt improve observable answer behavior when the retrieved context is fixed?

5.2 Indexes, Graph, and Runtime Settings

The dense/hybrid/graph experiment uses a FAISS index with 1,513,376 chunk vectors and a graph over the same chunk identifiers. The hybrid branch searches ten BM25 shards. Dense and hybrid retrieval use a 50-result search budget, broad filtering, and a final context of ten chunks; RRF uses $k = 60$ when BM25 is enabled. Graph variants take five seed chunks, traverse one hop, and retain at most ten graph-context chunks. The associated generation artifacts use gpt-4o-mini.

The reranker family fixes a 30-chunk RRF candidate pool (C_{30}) and a ten-chunk final context (T_{10}). It compares no reranking, RRF-only, mMiniLM, Qwen3, BGE, and Jina. The primary generator study compares GPT-4o-mini Base and CoT with the saved top-10 context fixed by question. The metadata study changes only the indexed string from chunk body to header-plus-chunk text; encoder, FAISS, top- $k = 10$, generator, prompt, seed, corpus, and benchmark remain fixed.

5.3 Benchmark and Metrics

The benchmark contains 500 questions: 400 answerable questions with gold chunk identifiers and 100 unanswerable questions. Retrieval scores use only the 400 answerable records. Given gold chunks G_i and retrieved top- k chunks $R_{i,k}$,

$$\text{Recall}@k_i = \frac{|G_i \cap R_{i,k}|}{|G_i|}, \quad (8)$$

$$\text{Hit}@k_i = \mathbb{1}[G_i \cap R_{i,k} \neq \emptyset].$$

MRR and nDCG reward an earlier gold rank. Candidate Recall@30 and Candidate Hit@30 define the evidence ceiling for the reranker family and are identical across its reranker conditions. Graph runs additionally report document- or identity-level precision because a graph policy can return fewer chunks.

For a normalized prediction y_i and reference r_i , exact match is

$$\text{EM}_i = \begin{cases} 1, & y_i = r_i, \\ 0, & y_i \neq r_i. \end{cases} \quad (9)$$

Token F1 uses token overlap O_i and prediction/reference lengths L_i^y, L_i^r :

$$P_i = \frac{O_i}{L_i^y}, \quad R_i = \frac{O_i}{L_i^r}, \quad F_{1,i} = \frac{2P_iR_i}{P_i + R_i}. \quad (10)$$

ROUGE-L uses the same F-score form with the token-level longest common subsequence in place of O_i . These overlap metrics are computed on successful answerable outputs; they are not legal-correctness scores.

The saved unanswerable score is a template-based answer/refuse diagnostic. Citation presence is the fraction of successful answerable outputs containing at least one marker, while structural coverage is the fraction of scored factual sentences with a marker. Neither measure tests claim-to-source entailment. Completion is successful rows divided by scheduled rows. Latency is reported in seconds as saved mean, p50, and p95 values where available; reranker runs use cached BM25 candidates and therefore do not represent full-online sparse latency.

5.4 Pairing, Uncertainty, and Claim Boundaries

Headline effects are paired by qa_id. Retrieval deltas use the shared answerable intersection; scheduled-denominator answerability diagnostics retain all 500 cases. Percentile paired-bootstrap confidence intervals use 10,000 resamples (seed 42) for headline metadata, dense/hybrid, reranker, and GPT comparisons, and 2,000 resamples for the smaller reranker slices. An interval crossing zero is reported as inconclusive rather than as evidence of an effect. Exact Match in the primary GPT pair also uses an exact McNemar test.

The primary GPT-4o-mini Base/CoT runs complete 500/500 cases and have identical ordered top-10 chunk-ID sequences for every question. DeepSeek, GLM, and Qwen prompt pairs are descriptive checks: four DeepSeek cases have different context order, and the Qwen manifests differ in code snapshot, timeouts, or retry limits. No prompt effect is pooled across models.

The artifacts contain no blinded legal-correctness labels, claim-level faithfulness labels, validated semantic judge, preferred answer/clarify/refuse labels, raw chain-of-thought, or reasoning-token accounting. Thus, no result is interpreted as proof of legal correctness, legal safety, citation entailment, or faithful reasoning merely because it improves a lexical, rank, citation-format, refusal-format, or latency metric.

5.4.1 Reproducibility ledger

Each condition is accompanied by a manifest, per-question output records, metric summary, and

Table 6: Frozen configurations used by the completed ablation families. A dash means that the component was disabled. Candidate/final values differ by family and are not pooled across rows.

Family	Configuration	Dense text	Sparse	Graph	Reranking	Candidate/final
Embedding	Chunk-only	chunk text	—	—	—	10/10
Embedding	Chunk+metadata	header + chunk	—	—	—	10/10
Retrieval/graph	DenseOnly	header + chunk	—	—	—	50/10
Retrieval/graph	HybridOnly	header + chunk	BM25	—	—	50/10
Retrieval/graph	DenseGraph	header + chunk	—	G-LRAG	—	50/10
Retrieval/graph	HybridGraph	header + chunk	BM25	G-LRAG	—	50/10
Reranker	RRF-only	shared RRF pool	BM25	—	RRF	30/10
Reranker	mMiniLM	shared RRF pool	BM25	—	mMiniLM CE	30/10
Reranker	Qwen3	shared RRF pool	BM25	—	Qwen3	30/10
Reranker	BGE	shared RRF pool	BM25	—	BGE-v2-m3	30/10
Reranker	Jina	shared RRF pool	BM25	—	Jina-v2	30/10
LLM reasoning	GPT-4o-mini Base	fixed context	—	—	Base prompt	fixed/10
LLM reasoning	GPT-4o-mini CoT	fixed context	—	—	CoT prompt	fixed/10

latency file. Derived analyses are versioned with their source artifacts under [ablation_study/](#) and [ablation_work/](#); the reranker candidate reconstruction and paired analysis are in [ablation_work/reranker_choices/analyze_reranker_metrics.py](#), while the generator post-hoc report is in [ablation_study/e2e_LLM_reasoning/results/report.md](#). The report reads these frozen files and does not call a model during analysis.

6 Ablation Studies

This chapter follows the ablation plan in [report_structure.md](#). We report five completed experiment families and retain the deterministic evidence available for each one. Comparisons are made within a family because candidate construction and context budgets differ; the results are not treated as one pooled factorial experiment.

6.1 Embedding Text Ablation: Chunk-Only versus Chunk+Metadata

This comparison holds the dense encoder, generator, benchmark, seed, and raw FAISS evaluation path fixed. Only the indexed text changes from the chunk body to a header-plus-chunk representation. On 400 paired answerable questions, metadata leaves exact chunk Recall@10 effectively unchanged (48.5% versus 48.5%) but increases document-level success from 65.8% to 70.2%. The paired document-success gain is +4.5 percentage points (95% CI [+0.5,+8.8]), whereas the MRR@10 change is −5.9 points (95% CI [−9.1, −2.7]). Metadata therefore improves document identification while making the first exact

gold chunk appear later; the complete scores are in [Table 7](#) and the retrieval curves and paired outcomes are shown in [Figure 2](#).

6.2 Retrieval Ablation: Dense versus Hybrid Retrieval

The dense/hybrid comparison uses the same header-plus-chunk dense index, 50-seed retrieval budget, 10-chunk final context, benchmark, and generator. Hybrid retrieval adds BM25 and reciprocal rank fusion. DenseOnly outperforms HybridOnly on Recall@10 (48.55% versus 42.57%) and nDCG@10 (0.2813 versus 0.2638); the full comparison is in [Table 8](#). The paired Recall@10 difference is −5.98 points with a 95% CI of [−9.71, −2.21]. Hybrid retrieval also raises recorded total median latency from 5.483 seconds to 31.538 seconds; paired uncertainty is reported in [Table 9](#).

6.3 Graph Ablation: With versus Without Graph Expansion

Graph variants expand five seed chunks by one hop and retain at most 10 chunks. Graph expansion currently displaces more evidence than it adds. Relative to DenseOnly, DenseGraph changes Recall@10 by −12.40 points and loses at least one previously retrieved gold chunk for 61 questions while adding a graph-only gold chunk for only one question. The HybridOnly to HybridGraph change is −11.81 points, with 59 questions losing a gold chunk and six gaining one; the displacement counts and rank deltas are summarized in [Table 10](#).

Graph contexts are denser because they contain fewer chunks on average (5.605 for DenseGraph and 6.305 for HybridGraph), and their identity pre-

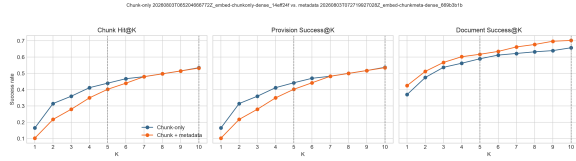


Figure 2: Embedding-text ablation diagnostics. The left panel shows success curves by retrieval depth; the right panel shows paired win, tie, and loss counts. Both plots use the shared answerable-question comparison.

Table 7: Embedding-text ablation on 400 paired answerable questions. Values are percentages; document success means that a retrieved chunk from the gold document is present in the indicated context. Deltas are chunk+metadata minus chunk-only.

Variant	Hit@5	Recall@5	MRR@5	Doc.@5	Hit@10	Recall@10	MRR@10	Doc.@10
Chunk-only	44.0	39.0	27.4	59.0	53.5	48.5	28.6	65.8
Chunk+metadata	40.2	36.8	20.9	61.8	53.2	48.5	22.7	70.2
Delta	−3.8	−2.2	−6.5	+2.8	−0.2	+0.1	−5.9	+4.5

cision is higher. This is a precision–coverage trade-off, not evidence of an overall retrieval-quality improvement.

6.4 Reranker Ablation

The reranker family compares no reranking, RRF-only, mMiniLM, Qwen3, BGE, and Jina over a reconstructed shared RRF pool of 30 candidates and a final top 10. All six runs cover the same 500 questions without retrieval or generation errors, and the reconstruction matches the recorded RRF-only top-10 prefix for 500/500 questions. Candidate Recall@30 is 0.5995 and Candidate Hit@30 is 0.6450 for every configuration, so the comparison measures ranking and retention within a fixed evidence ceiling.

Jina is the strongest retrieval configuration: relative to RRF-only, it improves Recall@10 by +9.23 points, MRR@10 by +13.35 points, and nDCG@10 by +11.85 points; all three 95% paired-bootstrap intervals exclude zero. Qwen3 and BGE also have positive intervals, while mMiniLM supplies a smaller positive gain at the lowest reranker cost. BGE has the highest token F1 and ROUGE-L, whereas Jina has the highest observed unanswerable accuracy. These answer metrics are lexical and abstention diagnostics, not legal correctness judgments. Aggregate scores and latency are linked in Table 11, with paired effects in Table 12 and diagnostic curves in Figure 3.

6.5 Generator Ablation: Base Prompt versus Chain-of-Thought Prompt

The generator-reasoning family keeps the intended retrieved top-10 context fixed for each question and changes the prompt mode. The primary comparison uses complete GPT-4o-mini Base and CoT runs: both cover all 500 benchmark questions and share 400 answerable question IDs. The frozen artifacts report zero retrieval-context mismatches. The DeepSeek-V3.1-Thinking, GLM-5, and Qwen3-8B pairs are complete but descriptive; four shared DeepSeek cases have different exact top-10 chunk-ID sequences, and the Qwen pair uses different code snapshots, timeout values, and retry limits.

On the 400 paired answerable questions, CoT decreases Token F1 from 24.72% to 23.15% (−1.57 points) and ROUGE-L from 21.60% to 19.90% (−1.71 points). Exact match is essentially unchanged (14.00% to 13.75%, −0.25 points). The Token F1 win/tie/loss count is 79/164/157 in favor of Base. The run inventory is in Table 13 and the paired GPT-4o-mini result is linked in Table 14.

6.6 Performance by Question Category and Answer Type

The largest Jina gains occur for legal-validity and citation questions, while the smaller extractive slice is inconclusive. The multi-hop slice has only 20 questions and is exploratory; hard and cross-document slices are too small for a main conclu-

Table 8: Dense versus hybrid retrieval. Retrieval values use 400 answerable questions; latency percentiles use all 500 cases.

Method	Recall@5	Recall@10	MRR@10	nDCG@10	Retrieval p50/p95	Total p50/p95
DenseOnly	0.3678	0.4855	0.2267	0.2813	0.797/0.856	5.483/9.946
HybridOnly	0.3140	0.4257	0.2252	0.2638	27.855/27.924	31.538/33.623

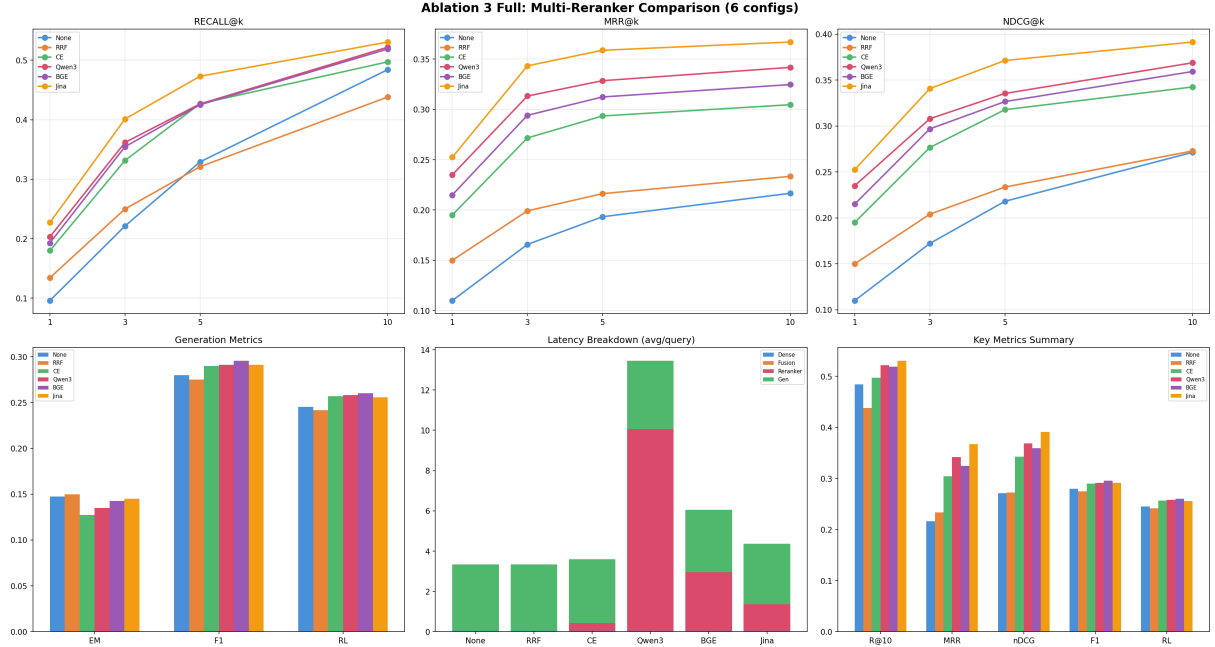


Figure 3: Reranker curves, answer-overlap diagnostics, latency breakdown, and headline metrics from the six-configuration run. The paired confidence intervals in Table 12 are the primary uncertainty analysis.

Table 9: Paired hybrid-minus-dense differences.

Metric	Mean delta	95% CI
Recall@10	-0.0598	[-0.0971, -0.0221]
MRR@10	-0.0015	[-0.0299, +0.0271]
nDCG@10	-0.0175	[-0.0437, +0.0085]

sion. The table reports Jina versus RRF-only deltas on answerable cases with 2,000 paired-bootstrap resamples; see Table 15.

6.7 Evidence Coverage, Abstention, and Citation Analysis

The GPT-4o-mini pair provides structural diagnostics for evidence availability, answerability decisions, and citations. These fields describe observable output format and saved parser coverage; they do not establish citation entailment, semantic refusal quality, or legal correctness. CoT lowers the false-refusal rate on answerable questions from 21.25% to 19.50%, while unanswerable recognition remains 98.00% for both prompts. Citation presence rises from 78.25% to 81.00% and mean

unique sources from 1.46 to 1.68, but structural citation coverage falls from 89.42% to 87.91%; the detailed diagnostics are in Table 16.

Evidence-stratified deltas do not reverse the result: CoT Token F1 is lower when all gold support is retrieved (-1.98 points, $n = 138$), when support is partial (-1.86 points, $n = 29$), and when support is absent (-1.30 points, $n = 233$). The saved records cannot support claim-level faithfulness.

6.8 Latency and Efficiency Comparison

Efficiency differences are substantial. Hybrid retrieval reports a one-time BM25 precomputation cost of about 13,495 seconds and a much larger per-query retrieval cost. Among rerankers, mMiniLM is fastest, Jina is intermediate, BGE is slower, and Qwen3 is much slower. The generator CoT prompt raises generation p50/p95 from 2.92/5.74 seconds to 3.68/6.17 seconds, and total p50/p95 from 13.99/17.21 to 14.74/18.42 seconds. Reranker p95, peak GPU memory, and controlled full-online throughput are unavailable. The com-

Table 10: Effect of graph expansion on top-10 gold evidence. Counts are over 400 answerable questions; graph-only gold is introduced by expansion and graph-missed gold was present before expansion but absent afterward.

Base	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Graph-only gold	Graph-missed gold	Avg. chunks	ID precision
DenseOnly $\rightarrow$ DenseGraph	-0.1240	-0.0222	-0.0439	1 query (0.25%)	61 queries (15.25%)	5.605	0.0800
HybridOnly $\rightarrow$ HybridGraph	-0.1181	-0.0240	-0.0411	6 queries (1.50%)	59 queries (14.75%)	6.305	0.0641

Table 11: Multi-reranker ablation over a shared 30-candidate pool and final top-10 context. Retrieval and overlap values are means over 400 answerable questions; recorded unanswerable accuracy and p50 latency use the complete 500-case runs.

Model	Recall@10	Hit@10	MRR@10	nDCG@10	Token F1	ROUGE-L	Unans. acc.	Rerank p50	Total p50
None	0.4842	0.5375	0.2167	0.2714	0.2801	0.2453	0.856	0.000	2.925
RRF-only	0.4382	0.4800	0.2335	0.2728	0.2750	0.2416	0.874	0.000	2.972
mMiniLM	0.4973	0.5425	0.3047	0.3425	0.2901	0.2568	0.872	0.385	3.343
Qwen3	0.5218	0.5725	0.3418	0.3688	0.2912	0.2583	0.876	9.695	13.007
BGE	0.5188	0.5675	0.3247	0.3593	0.2955	0.2603	0.868	2.872	5.848
Jina	0.5305	0.5850	0.3670	0.3914	0.2913	0.2558	0.884	1.278	4.158

plete stage and end-to-end comparison is linked in [Table 18](#).

6.9 Qualitative Ablation Examples and Failure Cases

The saved aggregate artifacts support a precise failure taxonomy even without hand-labelled legal case studies. Metadata has a document-provision trade-off: document-level success increases at top 10 while exact chunk MRR falls, so a retrieved document title is not proof that the governing provision is ranked early. Hybrid fusion can add lexical context while lowering exact gold-chunk coverage; a generation-overlap gain is not evidence that RRF preserved the gold ranked list. Graph expansion has a direct seed-displacement failure: it introduces a new top-10 gold chunk for only 1/400 Dense and 6/400 Hybrid questions, but removes at least one seed-retrieved gold chunk for 61 and 59 questions respectively. Finally, a fluent response with a structural citation marker is not a verified legally correct or entailed answer.

Future case studies should preserve the qa_id, gold chunk IDs, ordered candidate IDs before and after the changed component, serialized context, final answer, and citation parser output. A case must label whether its diagnosis is an automatic rank fact or a human legal judgement; this keeps illustrative examples auditable rather than anecdotal.

6.10 Summary of Ablation Findings

The completed evidence supports conditional engineering decisions. Metadata helps document-level retrieval but not exact chunk ranking. The tested hybrid configuration reduces exact-chunk cover-

age and adds latency. One-hop graph expansion increases context density while displacing more gold evidence than it adds. Cross-encoder reranking is the clearest positive retrieval result: Jina gives the strongest ranking scores, BGE gives the strongest lexical overlap, and mMiniLM gives the lowest recorded reranking cost. For the primary generator pair, CoT lowers lexical overlap under fixed context while improving only template-based decision diagnostics and adding latency. The consolidated recommendations are linked in [Table 19](#).

Table 12: Paired effects of each reranker relative to RRF-only. Deltas are in percentage points; intervals are 95% percentile bootstrap CIs from 10,000 question-level resamples over the 400 answerable cases. A negative false refusal delta indicates fewer refusals on answerable questions.

Reranker	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Δ false refusal
mMiniLM	+5.91 [+1.85, +9.96]	+7.12 [+3.55, +10.68]	+6.96 [+3.77, +10.26]	+0.25 [−3.00, +3.50]
Qwen3	+8.37 [+4.57, +12.25]	+10.84 [+7.24, +14.36]	+9.60 [+6.52, +12.73]	−0.25 [−3.25, +2.75]
BGE	+8.07 [+4.29, +11.80]	+9.12 [+5.50, +12.70]	+8.65 [+5.47, +11.84]	+0.75 [−2.25, +3.75]
Jina	+9.23 [+5.51, +12.98]	+13.35 [+9.51, +17.24]	+11.85 [+8.46, +15.27]	−1.25 [−4.00, +1.25]

Table 13: End-to-end generator-reasoning run inventory. Lexical metrics are means over successful answerable cases; decision accuracy is the saved template-based answerability diagnostic.

Model	Prompt	Run directory	Valid/500	EM (%)	Token F1 (%)	ROUGE-L (%)	Decision acc. (%)
GPT-4o-mini	Base	4o-mini-base	500/500	14.00	24.72	21.60	82.60
GPT-4o-mini	CoT	4o-mini-CoT	500/500	13.75	23.15	19.90	84.00
DeepSeek-V3.1-Thinking	Base	deepseek-base	500/500	22.25	17.84	15.28	92.00
DeepSeek-V3.1-Thinking	CoT	deepseek-CoT	500/500	22.25	16.51	14.12	92.20
GLM-5	Base	glm5-base	500/500	25.25	16.91	14.90	91.80
GLM-5	CoT	glm5-CoT	500/500	25.00	16.23	14.27	93.40
Qwen3-8B	Base	qwen3-8b-base	500/500	19.25	21.98	18.92	95.80
Qwen3-8B	CoT	qwen3-8b-CoT	500/500	17.25	20.05	17.17	95.20

Table 14: Primary paired GPT-4o-mini Base-versus-CoT comparison on 400 answerable questions. Deltas are CoT minus Base in percentage points; CIs are percentile bootstrap intervals from 10,000 paired resamples.

Metric	Base (%)	CoT (%)	Δ (pp)	95% paired CI (pp)
Exact Match	14.00	13.75	−0.25	[−1.50, +1.00]
Token F1	24.72	23.15	−1.57	[−2.24, −0.92]
ROUGE-L	21.60	19.90	−1.71	[−2.34, −1.10]

Table 15: Jina versus RRF-only on selected category and answer-type slices. Deltas are percentage points with 95% paired-bootstrap CIs.

Dimension	Slice	<i>n</i>	Δ Recall@10	Δ MRR@10	Δ nDCG@10
Category	citation	106	+11.89 [+2.82, +21.70]	+16.22 [+7.86, +24.51]	+15.82 [+7.51, +23.82]
Category	legal validity	91	+14.47 [+6.96, +21.43]	+18.90 [+10.28, +27.26]	+15.22 [+9.13, +21.50]
Category	single hop	172	+5.04 [−0.00, +10.37]	+9.63 [+4.34, +14.69]	+8.41 [+3.81, +13.05]
Category	multi hop	20	+7.50 [−0.83, +17.50]	+10.12 [−3.42, +24.33]	+7.95 [−0.77, +17.98]
Answer type	extractive	78	+0.13 [−8.85, +9.74]	+8.53 [−0.90, +17.39]	+7.14 [−1.33, +16.23]
Answer type	abstractive	199	+13.65 [+8.50, +18.80]	+14.74 [+9.61, +20.14]	+13.81 [+9.56, +18.71]
Answer type	boolean	123	+7.86 [+1.63, +14.50]	+14.18 [+7.64, +21.13]	+11.68 [+6.27, +17.95]

Table 16: Structural evidence, abstention, and citation diagnostics for the paired GPT-4o-mini runs.

Diagnostic	Base	CoT	Interpretation
Citation presence	78.25%	81.00%	at least one marker on answerable output
Structural coverage	89.42%	87.91%	saved factual-sentence marker share
Mean unique sources	1.46	1.68	structural source count
False refusal (answerable)	21.25%	19.50%	refusal-marker diagnostic
Unanswerable recognition	98.00%	98.00%	refusal-marker diagnostic

Table 17: Evidence-availability strata for the paired GPT-4o-mini reasoning comparison. Values are lexical means on the 400 paired answerable cases.

Evidence stratum	n	Base F1	CoT F1	Δ F1	Base ROUGE-L	CoT ROUGE-L	Δ ROUGE-L
Support fully retrieved	138	28.24	26.26	−1.98	25.64	23.34	−2.30
Partial support retrieved	29	25.62	23.76	−1.86	21.04	19.45	−1.59
No support retrieved	233	22.53	21.23	−1.30	19.29	17.92	−1.37

Table 18: Recorded latency comparisons. Values are p50/p95 seconds unless noted; reranker p95 was not recorded. Cached BM25 means reranker rows are not full-online sparse-latency measurements.

Configuration	Stage p50/p95	Total p50/p95	Stage
DenseOnly	0.797/0.856	5.483/9.946	retrieval
HybridOnly	27.855/27.924	31.538/33.623	retrieval
DenseGraph	0.798/0.860	4.998/9.750	retrieval + graph
HybridGraph	27.858/27.929	31.180/33.312	retrieval + graph
GPT-4o-mini Base	2.92/5.74	13.99/17.21	generation
GPT-4o-mini CoT	3.68/6.17	14.74/18.42	generation
mMiniLM	0.385/N/A	3.343/N/A	reranker
Qwen3	9.695/N/A	13.007/N/A	reranker
BGE	2.872/N/A	5.848/N/A	reranker
Jina	1.278/N/A	4.158/N/A	reranker

Table 19: Decision summary for the completed ablation families. Quality claims are limited to the deterministic metrics available in the saved artifacts.

Family	Changed component	Quality effect	Latency effect	Decision
Embedding	chunk text + metadata	document success +4.5 pp; MRR@10 −5.9 pp	not material in headline table	use when document identity is primary
Dense/hybrid	add BM25 + RRF	Recall@10 −5.98 pp	total p50 5.483 s → 31.538 s	do not use as default under current settings
Graph	one-hop expansion	Recall@10 −12.40 pp (Dense), −11.81 pp (Hybrid)	graph stage near dense retrieval cost	gate until seed-preserving expansion is tested
Reranker	reorder shared C30 pool	Jina: Recall +9.23 pp, MRR +13.35 pp, nDCG +11.85 pp	mMiniLM fastest; Jina p50 1.278 s	prefer Jina for retrieval quality; mMiniLM for latency
Generator	Base prompt → CoT	Token F1 −1.57 pp; ROUGE-L −1.71 pp	total p50 13.99 s → 14.74 s	retain Base for this lexical benchmark

7 Discussion, Limitations, and Ethical Considerations

7.1 Interpretation of the Main Findings

The ablations yield one clear positive retrieval intervention and several conditional or negative findings. Cross-encoder reranking improves the rank of evidence already admitted to the shared C30 pool; Jina has the strongest retrieval result, while BGE has the strongest lexical answer overlap (the paired evidence is linked in Table 12). By contrast, the tested hybrid and graph configurations reduce exact gold-chunk coverage, as shown by the dense/hybrid comparison in Table 8 and the graph displacement counts in Table 10. These statements concern a finite benchmark and its gold evidence, not legal advice quality in deployment.

7.2 Retrieval Quality, Context Coverage, and Latency Trade-offs

DenseOnly is the preferred retrieval baseline in the dense/hybrid study: it achieves Recall@10 of 0.4855 at 0.797-second retrieval p50. HybridOnly falls to 0.4257 and takes 27.855 seconds when amortized BM25 time is included (Table 8). Graph contexts appear more identity-precise because they contain only 5.605–6.305 chunks on average, but their reduced context size also removes gold coverage (Table 10). The reranker result has a different interpretation: it operates after a fixed candidate ceiling, so Jina’s gain is a ranking gain rather than proof that the underlying candidate retriever finds more evidence.

7.3 Implications for Legal RAG Design

For this benchmark, use header-plus-chunk embeddings when correct document identification is more valuable than the rank of an exact chunk; otherwise prefer the chunk-only representation (Table 7). Do not deploy the current BM25+RRF or one-hop graph setup as the default retrieval path without retuning. Prefer Jina when retrieval quality justifies its 1.278-second reranking p50, or mMiniLM when its lower 0.385-second cost is more important. Retain the Base prompt for lexical answer overlap. In all cases, present citations as sources to inspect and retain human review for legal interpretation.

7.4 Dataset and Coverage Limitations

The benchmark has 500 questions, of which only 400 are answerable by stored gold chunks. The hard stratum has one answerable item, while multi-hop has 20; those slices are descriptive and cannot establish stable subgroup effects. Gold chunk IDs are an evidence target, not necessarily the full set of legally acceptable authorities. Corpus coverage, language variation, temporal validity, and external legal references may all limit transfer beyond the data snapshot.

7.5 Knowledge Graph and Validity Limitations

The graph study evaluates a specific policy—five seeds, one hop, and ten retained chunks—rather than graph information in general. Its failures point to an engineering hypothesis: expansion substitutes for seeds too early. Future graph experiments should preserve high-ranked seeds, separately budget neighbors, verify relationship direction and validity semantics, and measure whether graph-only evidence is legally useful rather than merely novel.

7.6 Retrieval and Generation Limitations

The index, BM25 tokenization, RRF parameter, candidate budgets, rerankers, generator, prompt, and final context size are fixed settings, not optimized universally. Reranker timings use cached BM25 candidates and omit full-online sparse retrieval. Lexical overlap may reward reference phrasing rather than legal adequacy. Structural citations measure marker presence or parser coverage, not the relationship between a claim and the cited authority.

7.7 Evaluation Limitations

No blinded legal-correctness labels, claim-level faithfulness labels, or validated semantic judge are available. The report therefore does not claim that any configuration is legally more correct, safer, or more faithful solely because it improves F1, ROUGE-L, citation count, or refusal markers. Full-online p95 reranker latency, peak memory, and throughput are also absent. Non-GPT prompt pairs have provenance or control differences and remain descriptive.

7.8 Ethical Considerations and Safe Use

L_RAG should support research and source discovery, not replace a qualified legal professional. Users should inspect the retrieved authority, its date and validity, and the uncertainty of an answer. Systems should avoid unnecessary personal query data, should not obscure abstentions, and should treat well-cited generated text as a review cue rather than a legal conclusion.

8 Conclusion and Future Work

8.1 Summary of the Implemented System

L_RAG combines a Vietnamese legal chunk corpus, dense FAISS retrieval, optional BM25/RRF fusion, graph expansion, cross-encoder reranking, grounded generation, and saved per-question evaluation records. The ablation report isolates five component choices—embedding text, retrieval mode, graph policy, reranking, and generator prompting—using fixed question sets and paired metrics.

8.2 Main Contributions

The empirical contribution is a traceable set of controlled decisions: the embedded text representation changes document success but not exact chunk coverage; the tested hybrid and graph variants impose measurable coverage losses; reranking within a common pool produces substantial ranking gains; and CoT does not improve lexical answer overlap under fixed context. The report also separates retrieval evidence, answer overlap, refusal formatting, citation structure, and efficiency instead of collapsing them into one score.

8.3 Lessons Learned

The central lesson is that component labels are not outcomes. Metadata, hybrid retrieval, graph expansion, CoT, and cited output are not quality

guarantees. Candidate control and paired question-level deltas reveal effects obscured by aggregate scores: metadata can improve document discovery while delaying exact chunks, graph expansion increases density while removing evidence, and rerankers can only improve what has entered C30. Frozen manifests and complete row alignment are essential before interpreting a difference as causal.

8.4 Recommended System Configuration

For the evaluated benchmark, the recommended evidence path is dense retrieval with the Jina reranker when its latency is acceptable, a 10-chunk final context, and the Base prompt. Use the graph branch only after seed-preserving expansion is validated; do not select the current hybrid branch as a default. This is a benchmark-bounded configuration, not a deployment approval. A real deployment requires corpus-validity checks, source display, monitored latency, privacy controls, abstention handling, and legal expert review.

8.5 Future Improvements

Priority work is (1) blinded legal-expert annotation for correctness, completeness, and claim-to-citation entailment; (2) seed-preserving graph and neighbor-budget sweeps; (3) BM25/tokenization and RRF-weight tuning under full-online latency measurement; (4) candidate-pool sweeps beyond C30; and (5) temporal-validity and external-reference evaluation. These studies should retain per-case candidates and provenance so failures can be audited.

8.6 Final Conclusion

The ablation evidence supports a practical conclusion: improve the ordering of available legal evidence with reranking before adding unvalidated retrieval complexity. Jina is the strongest observed reranker in the fixed-pool study, while the current hybrid, graph, and CoT variants do not improve their primary quality measures enough to offset their costs or risks. The result is an evidence-based engineering recommendation, bounded by the benchmark and not a claim of legal correctness.

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A Dataset and JSONL Schemas

The corpus and evaluation artifacts use line-delimited JSON records so that a single question or passage can be audited without loading the full corpus. The minimum normalized document, provision, chunk, question, and run fields are listed in [Table 20](#).

The ground-truth object stores document, provision, and chunk IDs rather than only a text span. An unanswerable question has an empty gold set and remains in the scheduled denominator for answerability diagnostics. Prediction rows retain failed or skipped statuses in the ledger, while lexical metrics use only valid successful answerable rows. This distinction is required to reproduce the denominators in [Section 5](#).

B Example Documents, Provisions, Chunks, and Edges

The following compact example shows how one source is represented without including a full legal document:

```
document: doc_id=LAW-EXAMPLE-2024
           type=law, effective_date=2024-01-01
provision: provision_id=LAW-EXAMPLE-2024-A15
           article=15, parent=LAW-EXAMPLE-2024
chunk: chunk_id=LAW-EXAMPLE-2024-A15-02
       sequence=2, text="..."
edges: DOCUMENT_HAS_PROVISION -> A15
       PROVISION_HAS_CHUNK -> A15-02
       CHUNK_NEXT: A15-01 -> A15-02 -> A15-03
```

The visible passage can be shortened for a prompt, but its parent identifiers remain unchanged. A relation to an unavailable instrument is represented as an external stub rather than a fabricated document:

```
external_stub: stub_id=EXT-8841
               reference="Instrument X, Article 4"
relation: LAW-EXAMPLE-2024 --CITES--> EXT-8841
status: unresolved_target=true
       retrievable_text=false
```

These examples make the graph and retrieval joins explicit. They also explain why a high-ranked document can be counted as a document-level success while a specific article or chunk remains absent from the retrieved evidence.

Table 20: Core fields in the machine-readable report artifacts.

Record	Required identifiers	Representative fields
Document	<code>doc_id</code>	title, document type, issuing authority, issue/effective dates, validity status, raw source.
Provision	<code>provision_id, doc_id</code>	heading, article/clause/item path, sequence, parent provision.
Chunk	<code>chunk_id, provision_id, doc_id</code>	passage text, sequence, embedded text variant, reading-order pointers.
Question	<code>qa_id</code>	question, category, answer type, reference answer, ground-truth IDs.
Prediction	<code>qa_id, run_id</code>	answer, ordered context IDs, citation markers, refusal marker, status, error.
Run manifest	<code>run_id</code>	model/checkpoint, corpus snapshot, seed, configuration, code snapshot, timing files.

Table 21: Graph relation conventions.

Edge	Direction and use
DOCUMENT_HAS_PROVISION	Document to contained provision.
PROVISION_HAS_CHUNK	Provision to indexed chunk.
CHUNK_NEXT	Earlier chunk to next chunk in reading order.
PROVISION_NEXT	Earlier provision to next provision in reading order.
CITES	Citing instrument to cited instrument.
AMENDS/REPEALS	Later instrument to the instrument it changes or terminates.
REPLACES	Superseding instrument to the replaced instrument.

C Graph Relationship Vocabulary

Graph edges are directed and retain a source record, target record, relation type, provenance, and (when available) effective-date or confidence fields. The structural and legal relations used by L_RAG are summarized in Table 21.

Only verified directions are exposed to query-time traversal. An uncertain relation can remain in the audit ledger or point to an `ExternalStub`, but it is not silently promoted to an authoritative retrievable edge. The validity overlay also records unknown status explicitly; it does not infer current law from a similar title.

D Retrieval and Reranker Configurations

Table 22 records the model checkpoints, budgets, and scoring conventions used across the completed runs. It complements the family-level table in Section 5 by documenting implementation details that are useful when reproducing an individual run.

E Prompt and Output Formats

The Base and CoT conditions use the same retrieved context and output budget in the primary GPT-4o-mini comparison. The Base prompt asks for a concise answer supported by the supplied passages and provision-level citations. The CoT prompt adds an explicit reasoning instruction, but

the saved artifacts retain only the final answer and observable diagnostics, not hidden reasoning.

The output contract is conceptually:

```
answer_type: boolean | extractive |
              abstractive | unanswerable
answer: final answer text or INSUFFICIENT_CONTEXT
citations: [{doc_id, provision_id,
             chunk_id, marker}]
```

The parser records citation presence, the number of unique sources, structural coverage of scored factual sentences, invalid markers, and refusal markers. These fields measure formatting and parser behavior. They do not determine whether a cited provision entails the adjacent statement, whether an answer is legally complete, or whether abstention was substantively appropriate.

F Run Manifests and Additional Results

All paths below are relative to the repository root. The lean report commit retains manifests, aggregate metrics, derived reports, and latency summaries; large raw per-question outputs and retrieval caches remain local evaluation artifacts.

- Embedding-text comparison: [ablation_study/chunkOnly_vs_chunkMetadata/results/](#).
- Dense, hybrid, and graph comparison: [ablation_study/dense_vs_Hybrid_vs_Graph/results/](#).
- Reranker source and analysis: [ablation_work/reranker_choices/evaluation_runs/ablation3_reranker_v2/](#) and [ablation_work/reranker_choices/analyze_reranker_metrics.py](#).
- Generator-reasoning analysis: [ablation_study/e2e_LLM_reasoning/results/report.md](#) and [ablation_work/e2e_LLM_Reasoning/analyze_posthoc.py](#).
- The teammate writing allocation is recorded in [docs/report/work_share.md](#).

Table 22: System-wide component and model configuration reference.

Component	Model/checkpoint	Key settings	Role and convention
Dense retriever	intfloat/multilingual-e5-large	560M parameters; normalized inner product	Encodes chunk-only or title+header+chunk text in FAISS <code>IndexFlatIP</code> .
Sparse retriever	BM25 (rank_bm25)	$k_1 = 1.5$, $b = 0.75$; <code>underthesea</code>	Sharded lexical search over Vietnamese tokens.
Fusion	RRF	$k = 60$; C_{30} where enabled	Combines dense and BM25 ranks before reranking.
Graph overlay	G-LRAG	<code>as_of_date</code> ; one-hop test	Validity filtering and reading-order expansion with a ten-chunk cap.
mMiniLM	mmarco-mMiniLMv2-L12-H384-v1	117M; FP32	Lightweight cross-encoder baseline.
BGE	BAAI/bge-reranker-v2-m3	567M; multilingual	Cross-encoder with highest observed lexical overlap.
Jina	jinaai/jina-reranker-v2-base-multilingual	278M; position patch	Strongest observed fixed-pool retrieval scores.
Qwen3	Qwen/Qwen3-Reranker-0.6B	0.6B; FP16	CausalLM-based reranker with OOM retry handling.
Generator	gpt-4o-mini	$T = 0.0$; max 1024 tokens; T_{10}	Grounded answer, provision citation, or <code>INSUFFICIENT_CONTEXT</code> .

The detailed graph displacement counts, reranker paired intervals, category slices, evidence diagnostics, and latency table are linked in the relevant result sentences of Section 6. For example, the graph summary reports one graph-only gold recovery for DenseGraph versus 61 queries that lose a previously retrieved gold chunk, while HybridGraph reports six recoveries versus 59 losses. These values are not duplicated here so that the report has one canonical table for each result and one clickable path from the interpretive sentence to its evidence.